

# A NOTE ON THE STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE

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ABSTRACT. We specify a routine that calibrates the five canonical Heston [1993] parameters  $\Phi = (\theta, \kappa, \eta, \rho, v_0)$  to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot  $S_{\text{ref}}$ , the moneyness-normalised and grid-snapped strike  $K^*$ , the surface-coverage requirements, the configurable in-engine objective (defaulting to implied-volatility “IV-space” residuals), the IV-space restart selection and acceptance gate, and the boundary-pegging rejection rule.

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## 1. The model

Under the risk-neutral measure, with constant risk-free rate  $r$  and continuous dividend yield  $g$ , the log-price  $X_t = \ln S_t$  and instantaneous variance  $v_t$  of the SPX follow the Heston [1993] dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where  $W, B$  are independent Brownian motions, so  $\text{corr}(dX_t, dv_t) = \rho dt$ . The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with  $\theta > 0$  the long-run variance,  $\kappa > 0$  the mean-reversion speed,  $\eta > 0$  the volatility of variance (QuantLib's **sigma**),  $\rho \in [-1, 1]$  the leverage correlation, and  $v_0 > 0$  the initial variance. The Feller quantity [Feller, 1951]

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit;  $\mathcal{F}(\Phi) \geq 0$  guarantees  $v_t > 0$  almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

**1.1. Pricing operators.** Fix a valuation with maturity  $\tau > 0$  (in years), strike  $K$ , and type  $w \in \{+1, -1\}$  (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (5)$$

**Black price** [Black, 1976, Black and Scholes, 1973]. With total standard deviation  $\Sigma = \sigma\sqrt{\tau}$ ,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[ F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (6)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where  $\mathcal{N}$  is the standard normal CDF and  $F = F(S, \tau)$ .

**Heston price.** The Heston model price  $C_H(\Phi; S, K, \tau, w)$  is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (7)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for  $j \in \{1, 2\}$ ,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[ \frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (8)$$

and characteristic functions  $f_j = \exp(C_j + D_j v_0 + iu \ln F)$  given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[ (b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (9)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (10)$$

where  $\mu_1 = \frac{1}{2}$ ,  $\mu_2 = -\frac{1}{2}$ ,  $b_1 = \kappa - \rho\eta$ ,  $b_2 = \kappa$ . In code this integral is evaluated by the QuantLib [Ballabio and the QuantLib contributors] `AnalyticHestonEngine` (which uses a branch-stable formulation of (8)); equations (7)–(8) define the operator it realises.

**Implied volatility.** The Black-implied volatility of a price  $C$  is the inverse of (6) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (11)$$

computed numerically (QuantLib `blackFormulaImpliedStdDev` on the forward (5), returning  $\Sigma$ , then  $\sigma = \Sigma/\sqrt{\tau}$ ). This operator defines the IV-space residuals of §5.

## 2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day  $d$ , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by  $i$ , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (12)$$

being respectively the quote timestamp, strike, type  $w_i \in \{+1, -1\}$ , traded size, trade price, trade-implied volatility, the underlying level *at trade time*  $t_i$ , and the days to maturity. Maturities enter the engine as  $\tau_i$  in years under Actual/365, i.e.  $T_i = \tau_i/365$ .

The day is rate-stamped with  $r = r(d)$  and  $g = g(d)$ , the most recent risk-free and dividend rates on or before  $d$  (a flat-forward curve for each; under flat curves the evaluation date is immaterial to  $T_i$ , so it is set to the QuantLib “today”).

**Pre-filter.** We keep only economically usable, Heston-appropriate quotes. Write the ratio moneyness  $\varrho_i = e^{-|\ln(K_i/S_i)|} \in (0, 1]$  (it equals 1 at the money and falls as the strike moves out). The kept set is

$$\mathcal{D} = \{ i : \sigma_i > 0, \tau_{\min} \leq \tau_i \leq \tau_{\max}, \underline{m} < \varrho_i < \bar{m} \}. \quad (13)$$

The maturity floor  $\tau_{\min}$  (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive  $\eta, \kappa$  to Feller-violating extremes that pollute the pooled fit; the ceiling  $\tau_{\max}$  drops thin, stale long tenors. The moneyness band keeps only out-of-the-money quotes: the cutoff  $\bar{m}$  removes the near-ATM rows, and the floor  $\underline{m}$  removes the deep-OTM lottery-ticket tail ( $\varrho_i < \underline{m}$ , roughly  $|\ln(K_i/S_i)| > 0.51$ ), whose extreme prices peg the fit to its bounds. Flooring that tail recovers acceptance and tightens the fit while keeping the full tradeable wing.

## 3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot  $S$ , yet the trades in  $\mathcal{D}$  occur across an intraday range of underlying levels  $\{S_i\}$ . We reconcile the two with one reference spot and a moneyness re-striking.

**Reference spot.** Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (14)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH\_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (15)$$

A flagged day ( $R > R_{\max}$ ) is still calibrated and written, but the normalisation below is recorded as strained (its  $S_{\text{ref}}$  should be treated with suspicion).

**Moneyness normalisation and grid snap.** Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width  $\Delta$ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (16)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in  $K/S$  rather than in the absolute strike. Re-striking by (16) therefore re-centres the whole day onto  $S_{\text{ref}}$  so that trades executed at different  $S_i$  but the *same* moneyness share a common surface column. The grid snap collapses the continuum of  $K_i^*$  values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract's *original* ( $S_i, K_i$ ), since the calibrated  $\Phi$  is spot-independent.

#### 4. Surface construction

From the normalised trades we build a partial  $K^* \times \tau$  implied-volatility surface, selecting the most liquid, nearest-the-money quotes.

**Maturity selection.** For each distinct maturity  $\tau$  present in  $\mathcal{D}$ , let  $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$  be its traded volume. Keep the  $N_t$  highest-volume maturities,

$$\mathcal{T} = \text{top-}N_t \text{ maturities by } V(\tau). \quad (17)$$

**Strike selection (per maturity, per wing).** Within each kept maturity  $\tau \in \mathcal{T}$ , separate calls ( $w_i = +1$ , OTM with  $K_i^* \gtrsim S_{\text{ref}}$ ) from puts ( $w_i = -1$ , OTM with  $K_i^* \lesssim S_{\text{ref}}$ ) and keep the  $n_k$  nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{n_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{n_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (18)$$

**Pivot to a surface.** For each retained cell  $(k, \tau)$  with  $k \in \mathcal{K}_\tau$ , set the surface entry to the implied volatility of the *latest* trade in that cell,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (19)$$

and leave it missing otherwise. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (20)$$

be the set of populated cells; these become the calibration targets (helpers).

**Coverage requirement.** To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (21)$$

A day failing (21) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

### 5. The optimization problem

Let  $\mathcal{N}$  index the targets with market data  $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$ , all priced at the single spot  $S_{\text{ref}}$ . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^5 [\ell_j, h_j] \subset \mathbb{R}^5, \quad (\ell, h) \text{ in the } (\theta, \kappa, \eta, \rho, v_0) \text{ order of Table 2.} \quad (22)$$

**5.1. Residuals.** Two residual systems are defined: an IV-space system and a price-space system. The in-engine objective is configurable. It defaults to the IV-space system (26) below, and can instead be set to the price-space system (23). The choice changes only what each restart minimises. Selection (30) and the acceptance gate (32) always run on the IV-space error  $J_{\text{iv}}$ , so under the default objective the optimizer minimises the very error that then selects and gates.

**Price-space (alternative in-engine objective).** For each target the market value is the Black price at the quoted vol,  $C_n^{\text{mkt}} = C_{\text{B}}(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$ , and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Phi) = \frac{C_{\text{H}}(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (23)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Phi) = \left( \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Phi)^2 \right)^{1/2}. \quad (24)$$

The per-target residual  $\varepsilon_n^{\text{px}}$  is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt [Marquardt, 1963] is a sum-of-squares method, so it drives the squared residual norm  $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Phi)^2$  to a local minimum (equivalently  $J_{\text{px}}$ , with which it shares a minimiser).

**IV-space (default in-engine objective; selection and gate).** Invert each Heston price back to a Black vol via (11) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Phi) &= \mathcal{I}(C_{\text{H}}(\Phi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Phi) &= \sigma_n^{\text{mod}}(\Phi) - \hat{\sigma}_n, \end{aligned} \quad (25)$$

with objective

$$J_{\text{iv}}(\Phi) = \left( \frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Phi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Phi) \text{ inverts finitely}\}. \quad (26)$$

$J_{\text{iv}}$  is measured in volatility points; it is the natural surface-fit metric and, unlike  $J_{\text{px}}$ , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

**5.2. Multi-start local optimization.** A single start into a non-convex problem is fragile [Mikhailov and Nögel, 2003], so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip}\left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau}\right)^2, [10^{-3}, 0.25]\right), \quad (27)$$

and define the seed set  $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$  of  $(v_0, \kappa, \theta, \eta, \rho)$  tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (28)$$

For each seed  $s \in \mathcal{S}$  we solve the box-constrained nonlinear least-squares problem

$$\hat{\Phi}_s = \arg \min_{\Phi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{obj}}(\Phi)^2 \quad (\text{LM, started at } \Phi^{(0)} = s), \quad (29)$$

using Levenberg–Marquardt under the box constraint  $\mathcal{B}$  (`QuantLib NonhomogeneousBoundaryConstraint`). The per-target residual  $\varepsilon_n^{\text{obj}}$  is the configured in-engine objective: by default the IV-space residual  $\varepsilon_n^{\text{iv}}$  (25), optionally the price-space residual  $\varepsilon_n^{\text{px}}$  (23). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Phi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Phi}_s). \quad (30)$$

Ranking by  $J_{\text{iv}}$  (not  $J_{\text{px}}$ ) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

## 6. Acceptance gate

$\Phi^*$  is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction  $\delta$ , be

$$\text{PEGGED}(\Phi) = \bigvee_{j=1}^5 \left[ (\Phi_j - \ell_j) < \delta (h_j - \ell_j) \quad \vee \quad (h_j - \Phi_j) < \delta (h_j - \ell_j) \right]. \quad (31)$$

The day is accepted iff

$$\boxed{J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Phi^*)}. \quad (32)$$

The first clause is the IV-space gate ( $\varepsilon_{\text{iv}} \approx 2$  vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of  $J_{\text{iv}}$ . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

**Why an IV-space gate.** An earlier version gated on  $J_{\text{px}}$  directly. On a pooled day rich in deep-OTM wings,  $J_{\text{px}}$  is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with  $J_{\text{iv}}$  aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is  $\approx 0.8$ – $1.4$  vol points, and the residual rejections are boundary-pegged  $\kappa$  or  $\rho$  (a separate identification lever), not fit-quality failures.

## 7. Output and diagnostics

An accepted day yields one parameter row keyed by date,

$$(d, S_{\text{ref}}, r, g, \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*, \mathcal{F}(\Phi^*), J_{\text{iv}}, J_{\text{px}}, \dots, R, \text{HIGH\_MOVE}), \quad (33)$$

together with coverage counts ( $|\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|$ ) and total volume. Because the calibration is once-daily, these rows accumulate into a single file, regenerated

each run from the accepted days. For repricing diagnostics each retained contract is valued under  $\Phi^*$  at its *original*  $(S_i, K_i)$ , not  $(S_{\text{ref}}, K^*)$ , since the structural parameters are spot-independent; this gives an honest comparison against the actual trade price  $P_i$ .

## 8. Constants

| Symbol                    | Code name             | Value     | Role                                    |
|---------------------------|-----------------------|-----------|-----------------------------------------|
| $\tau_{\min}$             | MIN_DTM               | 14 days   | minimum maturity                        |
| $\tau_{\max}$             | MAX_DTM               | 730 days  | maximum maturity                        |
| $\underline{m}$           | OTM_MONEYNESSE_FLOOR  | 0.6       | OTM ratio-moneyness band                |
| $\bar{m}$                 | OTM_MONEYNESSE_CUTOFF | 0.6       | OTM ratio-moneyness band                |
| $R_{\max}$                | MAX_MOVE_PCT          | 0.03      | intraday-range flag threshold           |
| $\Delta$                  | STRIKE_GRID           | 5.0       | SPX strike lattice for $K^*$ snap       |
| $N_t$                     | MAX_NT                | 20        | maturities kept (by volume)             |
| $n_k$                     | MAX_NK                | 40        | strikes kept per wing                   |
| $M_t$                     | MIN_MATS              | 3         | minimum maturities in surface           |
| $M_k$                     | MIN_STRIKES           | 5         | minimum strikes in surface              |
| $M_{\text{cell}}$         | MIN_CELLS             | 12        | minimum populated cells $ \mathcal{N} $ |
| $\varepsilon_{\text{iv}}$ | IV_RMSE_ACCEPT        | 0.02      | IV-space acceptance gate (vol pts)      |
| $\delta$                  | BOUND_TOL             | $10^{-3}$ | boundary-pegging tolerance fraction     |

TABLE 1. Surface, selection, and gate constants.

|              | $\theta$  | $\kappa$  | $\eta$    | $\rho$ | $v_0$     |
|--------------|-----------|-----------|-----------|--------|-----------|
| lower $\ell$ | $10^{-4}$ | $10^{-2}$ | $10^{-2}$ | -0.999 | $10^{-4}$ |
| upper $h$    | 1.0       | 20.0      | 2.0       | 0.5    | 1.0       |

TABLE 2. Box bounds  $\mathcal{B}$ , in the `model.params()` order  $(\theta, \kappa, \eta, \rho, v_0)$ . The  $\rho$  upper bound is kept slightly positive (equity leverage implies  $\rho < 0$ ) and  $\eta$  is capped at 2.0 (SPX vol-of-vol  $\sim 0.3$ – $1.2$ ).

## 9. Empirical findings

We ran the routine over the full CBOE SPX trade history available to us, 3,217 trading days from 3 January 2012 to 15 October 2024. The figures below are for the default IV-space objective (26). The price-space objective (24) is the configurable alternative, discussed at the end of the section. Of these days, 1,631 (50.7%) passed the acceptance gate (32) and contribute a parameter row; the 1,586 rejected days are logged with their cause. The rejection split is dominated by boundary pegging: 1,467 days (92.5% of rejections) are rejected for a pegged parameter, 116 for an IV-space miss ( $J_{\text{iv}} > \varepsilon_{\text{iv}}$ ), and 3 for having no usable trades. Pegging (almost always  $\kappa \rightarrow 20$  or  $\rho \rightarrow -0.999$  hitting the box wall) is therefore the binding constraint on the accept rate, not surface quality.

**Fit quality.** On the accepted set the surface fit is excellent. The IV-space error (26) has a median of 0.80 vol points (mean 0.84, 90th percentile 1.15, maximum 1.99), so  $J_{\text{iv}}$  sits inside the 2-vol-point gate on every accepted day and the gate never binds in practice. Acceptance is governed by the no-pegging clause alone. An independent repricing check (each contract valued under  $\Phi^*$  at its *original*  $(S_i, K_i)$  and compared to the trade price) gives a per-day median relative price

error of 11.4% with a 90th percentile of 17.4%. Those figures are inflated by the deep-OTM wings, whose near-zero prices make any absolute miss a large relative one. This is exactly the effect that motivated gating in IV space rather than price space (§6). A separate validator clears 1,623 of 1,631 accepted days (99.5%); the 8 hard failures are repricing-fit outliers, not parameter pathologies.

**Parameter levels.** The accepted parameters are economically sensible and stable across the 2012–2024 sample. The leverage correlation is consistently and strongly negative, median  $\rho = -0.74$  (10th–90th percentile  $[-0.81, -0.61]$ ), as equity-index smiles require. Long-run variance median  $\theta = 0.067$  ( $\approx 25.8\%$  annualised vol) and initial variance median  $v_0 = 0.026$  ( $\approx 16.2\%$ ) bracket realistic SPX levels, with  $v_0$  the lower of the two on the typical day. Mean reversion median  $\kappa = 2.37$  sits in the usual SPX range. The vol-of-vol is high, median  $\eta = 1.34$  (10th–90th percentile  $[0.79, 1.84]$ ). Because the gate rejects any boundary-pegged fit, the accepted set is pegging-free by construction: no accepted day carries  $\rho$  below  $-0.99$  or reaches the  $\kappa$  ceiling. A clean interior set is what survives the gate, not evidence that pegging is solved on the underlying population.

**Feller condition is the standout caveat.** The gate does not test Feller, and these fits violate it without exception:  $\mathcal{F}(\Phi^*) = 2\kappa\theta - \eta^2 < 0$  on all 1,631 accepted days, with median  $\mathcal{F} = -1.44$ . The driver of the violation is the large vol-of-vol relative to  $\kappa\theta$ .  $\eta$  exceeds 1.5 on 596 days (36.5%) and approaches its 2.0 cap (max 1.997). A pinned  $v_0$  or a degenerate  $\kappa$  are rare, so among the validator’s soft flags Feller violation and an elevated  $\eta$  are by far the most common. Together they mark these fits as ones whose variance path can reach zero, even where the surface fit itself is tight. The wide surface coverage is what lifts  $\eta$  here: putting the full OTM wings into the fit demands more vol-of-vol than a near-money fit, so both  $\eta$  and the Feller violation run higher than under the narrower near-money calibration of earlier drafts.

**Comparison with the earlier price-space baseline.** An earlier draft reported the price-space objective on a narrower surface (near-money strikes only, a higher maturity floor) as the default. That run accepted 1,713 of 3,215 days (53.3%) with a lower vol-of-vol (median  $\eta \approx 0.89$ ,  $\eta > 1.5$  on 8.5% of days) and Feller violated on 99.3%. Two things changed the picture here. The default objective is now the IV-space residual, and the surface coverage was widened to carry the full tradeable wing and longer maturities, with a deep-OTM floor that drops the lottery-ticket tail. The widening is the larger effect. It removes what looked like a systematic wing underfit: with the wings in the fit, the  $|\ln(K/S)|$  residual is small and mixed in sign (overall RMSE 1.1 vol points), so the near-money-only calibration was extrapolating, not capturing an in-sample bias. The same widening raises  $\eta$  and pushes Feller violation to every accepted day. The accept rate is little changed (50.7% against 53.3%), and pegging stays the dominant rejection cause under either setup.

**Price-space as the alternative objective.** Switching the in-engine objective back to price-space (24) changes only what each restart minimises, not how a day is selected or gated (§5). The IV-space default weights the deep-OTM wings evenly in vol points, whereas the price objective discounts them by their small prices. One consequence is specific to the IV-space default: it no longer controls relative price, so the returned price RMSE is large on the cheap wing

cells (maximum  $J_{\text{px}} = 120.8$ , with 656 days above 0.2). Those days are tight in vol points yet far off in relative price on a near-worthless wing cell, which is why the gate runs in IV space.

**9.1. Example option prices.** Produced by the market<sup>1</sup> calibrated pricing operator  $C_H(\Phi^*)$  (7) (32).

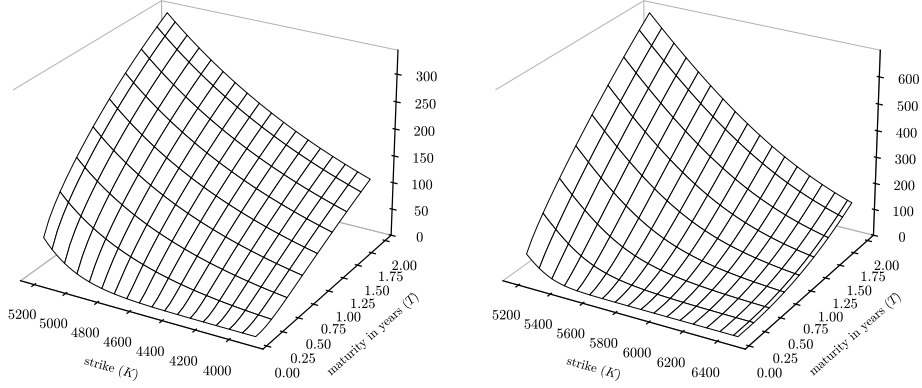


FIGURE 1. Heston OTM option prices for  $S_{\text{ref}} 5194.443$  on April 05, 2024 with  $\Phi^* = (0.0725, 0.7148, 0.666, -0.6408, 0.0198)$ : puts wing (left) and calls wing (right).

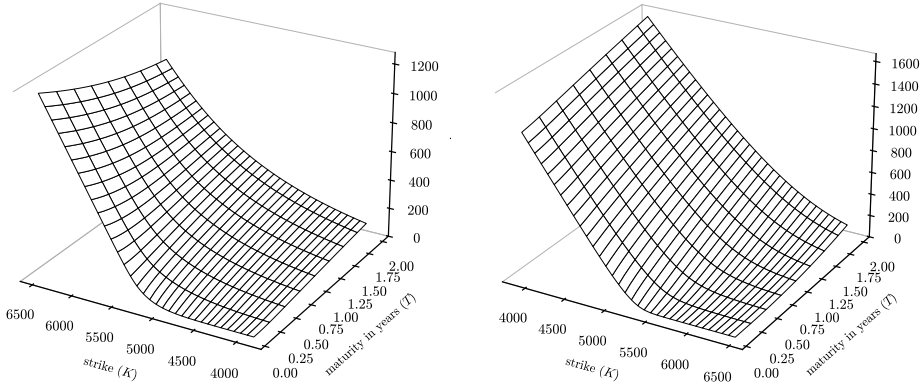


FIGURE 2. All put (left) and call (right) options from Figure 1

**9.2. Partial conclusions.** The objective is not the binding choice. Pegging is a structural degeneracy in the  $(\kappa, \rho, \eta)$  skew subspace, and it survives a change of residual. The pooled SPX surface demands more skew than single-factor Heston can supply without driving one of  $\kappa$  or  $\rho$  to a corner, and the optimiser picks

<sup>1</sup>The fit uses 1467 calibration cells across 20 maturities and 313 strikes, backed by 374,209 contracts of traded volume. The reference spot was  $S_{\text{ref}} = 5194.443$ , priced under a risk-free rate of 4.99% and a dividend rate of 1.39%. The intraday spot range was 1.46%. Fit quality is 0.33 vol points of implied-volatility RMSE, with a relative-price RMSE of 0.0605. The Feller condition violates at this calibration, with  $2\kappa\theta - \eta^2 = -0.3398$ .

whichever corner. Acceptance therefore turns on the no-pegging clause, not on fit quality or on which residual is minimised. The levers that would move it are the identification fixes: anchoring  $\kappa$ , a soft Feller penalty, and liquidity weighting.

**9.3. Avenues for improvement.** Acceptance is near one half of attempted days, and Feller violation is now universal on the accepted set. The maturity floor has already been raised to  $\tau_{\min} = 14$  days and an objective wing-weighting has been wired but tested null; neither moved acceptance. The remaining levers attack the  $(\kappa, \rho, \eta)$  degeneracy at its source. Fixing or anchoring  $\kappa$  from a cross-day-robust estimate would break the  $\kappa \leftrightarrow \rho$  trade-off so  $\rho$  stops pegging. A soft Feller penalty in the restart-selection criterion, with a revisit of the  $\eta$  cap, would curb the now-universal Feller violation. Warm-starting each day from the previous day's accepted parameters would stabilise the search. The Levenberg–Marquardt end criteria have not been tuned and are a further candidate.

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### Appendix A. Algorithm

Algorithm 1 states the per-day routine; the driver applies it to every trading day and accumulates accepted rows into a single file (§7).

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**Algorithm 1** Per-day Heston calibration

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**Require:** day  $d$  of OTM trades; rates  $r(d), g(d)$ ; constants of Table 1

- 1:  $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$  (pre-filter (13))
- 2: **if**  $r(d)$  or  $g(d)$  undefined on/before  $d$  **then**
- 3:   **return**  $\emptyset$  (no rate)
- 4: **end if**
- 5:  $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$ ;  
 $R \leftarrow \max_i S_i / \min_i S_i - 1$ ;  
 $\text{HIGH\_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6:  $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$  with  $m_i = K_i / S_i$  (re-strike (16))
- 7:  $\mathcal{T} \leftarrow$  top- $N_t$  maturities by volume; build  $\mathcal{K}_\tau$  ( $n_k$  per wing);  
pivot to  $\{\hat{\sigma}_{k,\tau}\}$ , cells  $\mathcal{N}$
- 8: **if** coverage (21) fails **then**
- 9:   clear stale diagnostics; **return**  $\emptyset$
- 10: **end if**
- 11:  $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$ ; build seeds  $\mathcal{S}$  (28)
- 12: **for**  $s \in \mathcal{S}$  **do**
- 13:    $\hat{\Phi}_s \leftarrow$  LM minimiser of  $\sum_n (\varepsilon_n^{\text{obj}})^2$  on  $\mathcal{B}$  from  $s$  (29)
- 14:   compute  $J_{\text{iv}}(\hat{\Phi}_s)$  (26)
- 15: **end for**
- 16:  $\Phi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Phi}_s)$  (select (30))
- 17: **if**  $J_{\text{iv}}(\Phi^*) \leq \varepsilon_{\text{iv}}$  **and**  $\neg \text{PEGGED}(\Phi^*)$  **then**
- 18:   write repriced diagnostics; **return**  $\text{row}(d, S_{\text{ref}}, r, g, \Phi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20:   clear stale diagnostics; **return**  $\emptyset$  (rejected)
- 21: **end if**

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## Appendix B. Best fits

